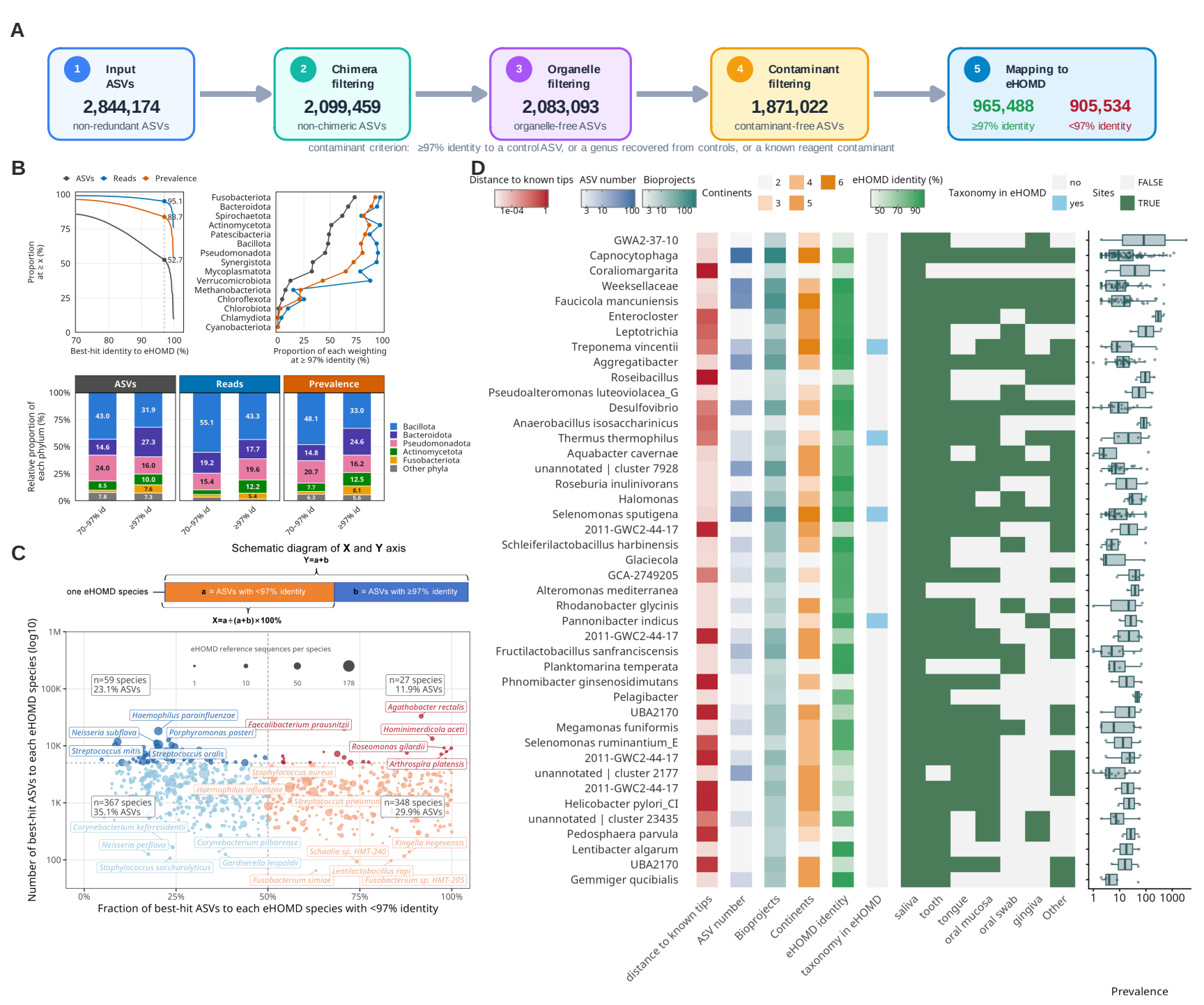

A
1 Input ASVs
2,844,174
non-redundant ASVs
2 Chimera filtering
2,099,459
non-chimeric ASVs
3 Organelle filtering
2,083,093
organelle-free ASVs
4 Contaminant filtering
1,871,022
contaminant-free ASVs
5 Mapping to eHOMD
965,488
≥97% identity
905,534
<97% identity
contaminant criterion: ≥97% identity to a control ASV, or a genus recovered from controls, or a known reagent contaminant
B
ASVs
Reads
Prevalence
Proportion at ≥ X (%)
Best-hit identity to eHOMD (%)
Fusobacteriota
Bacteroidota
Spirochaetota
Actinomycetota
Patescibacteria
Bacillota
Pseudomonadota
Synergistota
Mycoplasmatota
Verrucomicrobiota
Methanobacteriota
Chloroflexota
Chlorobiota
Chlamydiota
Cyanobacteriota
Proportion of each weighting at ≥ 97% identity (%)
Relative proportion of each phylum (%)
70–97% id
≥97% id
Bacillota
Bacteroidota
Pseudomonadota
Actinomycetota
Fusobacteriota
Other phyla
C
Schematic diagram of X and Y axis
Y=a+b
one eHOMD species
a = ASVs with <97% identity
b = ASVs with ≥97% identity
X=a÷(a+b)×100%
eHOMD reference sequences per species
n=59 species 23.1% ASVs
n=27 species 11.9% ASVs
n=367 species 35.1% ASVs
n=348 species 29.9% ASVs
Haemophilus parainfluenzae
Neisseria subflava
Porphyromonas pasteri
Streptococcus mitis
Streptococcus oralis
Faecalibacterium prausnitzii
Agathobacter rectalis
Hominimerdicola aceti
Roseomonas gilardii
Arthrospira platensis
Staphylococcus aureus
Haemophilus influenzae
Streptococcus pneumon
Corynebacterium kefirresidentii
Neisseria perflava
Corynebacterium pilbarense
Schaalia sp. HMT-240
Kingella negevensis
Staphylococcus saccharolyticus
Gardnerella leopoldii
Lentilactobacillus rapi
Fusobacterium simiae
Fusobacterium sp. HMT-205
Number of best-hit ASVs to each eHOMD species (log10)
Fraction of best-hit ASVs to each eHOMD species with <97% identity
D
Distance to known tips
ASV number
Bioprojects
Continents
eHOMD identity (%)
Taxonomy in eHOMD
Sites
GWA2-37-10
Capnocytophaga
Coraliomargarita
Weeksellaceae
Faucicola mancuniensis
Enterocloster
Leptotrichia
Treponema vincentii
Aggregatibacter
Roseibacillus
Pseudoalteromonas luteoviolacea_G
Desulfovibrio
Anaerobacillus isosaccharinicus
Thermus thermophilus
Aquabacter cavernae
unannotated | cluster 7928
Roseburia inulinivorans
Halomonas
Selenomonas sputigena
2011-GWC2-44-17
Schleiferilactobacillus harbinensis
Glaciecola
GCA-2749205
Alteromonas mediterranea
Rhodanobacter glycinis
Pannonibacter indicus
2011-GWC2-44-17
Fructilactobacillus sanfranciscensis
Planktomarina temperata
Phnomibacter ginsenosidimutans
Pelagibacter
UBA2170
Megamonas funiformis
Selenomonas ruminantium_E
2011-GWC2-44-17
unannotated | cluster 2177
2011-GWC2-44-17
Helicobacter pylori_CI
unannotated | cluster 23435
Pedosphaera parvula
Lentibacter algarum
UBA2170
Gemmiger qucibialis
distance to known tips
ASV number
Bioprojects
Continents
eHOMD identity
taxonomy in eHOMD
saliva
tooth
tongue
oral mucosa
oral swab
gingiva
Other
Prevalence